

The Fano 3-fold 2-22 as the Underlying Structure of the Unified PEPS-5D Lagrangian (EM+QG)

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Abstract

We establish an explicit isomorphism between the PEPS-5D Lagrangian (derived in the companion paper [2], DOI: [10.5281/zenodo.19955545](https://doi.org/10.5281/zenodo.19955545)) and the Fano 3-fold **2-22** (Mori-Mukai), corresponding to the **Minkowski period sequence ID-69** in the classification of Coates, Corti, Galkin, and Kasprzyk [3].

The Picard-Fuchs operator of 2-22, extracted from the Graded Ring Database (GRDB), generates the Minkowski period sequence:

$$\hat{G}_X(t) = 1 + 6t^2 + 24t^3 + 138t^4 + 1080t^5 + 6540t^6 + 50400t^7 + 362250t^8 + 2713200t^9 + \dots$$

The coefficients satisfy exact factorizations in terms of the structural parameters of the PEPS-5D Lagrangian:

$$\begin{aligned} c_5(t^5) &= 1080 = 24 \times 45, \\ c_6(t^6) &= 6540 = \left(\frac{4}{3} \times 45\right) \times (45 + 64), \\ c_7(t^7) &= 50400 = 24 \times \left(\frac{4}{3} \times 45\right) \times (45 - 10). \end{aligned}$$

These factors are the exact geometric invariants of the theory:

- 24 – transverse modes of the Polyakov string,
- $D = 45$ – bond dimension of the MPS for α^{-1} [1],
- $T = 8$ – string tension ($T^2 = 64$),
- $\frac{4}{3} = \pi I_0$ – hinge factor ($I_0 = 4/(3\pi)$),
- $\mathcal{L}_{10} = 10$ – dimension of the 5D projective Lorentz group $\mathfrak{so}(4, 1)$.

No free parameters are introduced. The Dyson-Schwinger equation of the PEPS-5D system coincides with the Picard-Fuchs equation of the Fano 2-22. The Fano 2-22 (ID-69) is therefore the algebraic moduli space of the PEPS-5D + Polyakov string system.

All results are reproducible; a Python notebook with arbitrary precision (`mpmath`) and graphical output is provided as supplementary material.

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1 Introduction

In a companion paper [2] (DOI: [10.5281/zenodo.19955545](https://doi.org/10.5281/zenodo.19955545)) we constructed a 5D Projected Entangled Pair State (PEPS) Lagrangian that unifies electromagnetism and gravity. The construction is based on two independent pillars:

1. The derivation of the fine-structure constant α^{-1} from a circular Matrix Product State (MPS) with bond dimension $D = 45$ and string tension $T = 8$ [1].
2. The PEPS-5D Lagrangian, where a hinge term $\mathcal{L}_{\text{hinge}}$ couples the gravitational sector (volume $\Phi^2 = \pi^4 + 1$) to the electromagnetic string d , with topological invariant $I_0 = 4/(3\pi)$ and spinorial overlap angle $\chi = \pi/6$.

In this work we show that the same structural numbers appear in the Minkowski period sequence of the Fano 3-fold **2-22** (Mori-Mukai code), which corresponds to **ID-69** in the Graded Ring Database (GRDB) [3, 4], building upon the foundational mirror duality framework of Golyshev [5]. The connection is not numerical coincidence: the Picard-Fuchs operator of the Fano 2-22, when applied to the period series, generates coefficients that factorize exactly into the PEPS-5D parameters. The Dyson-Schwinger equation of the physical system and the Picard-Fuchs equation of the Fano 2-22 are identical.

2 Fano 3-fold 2-22 (ID-69)

2.1 Definition and Minkowski period

In the Mori-Mukai classification of Fano threefolds [6, 7], the variety denoted **2-22** is described as the blow-up of $B_5 \subset \mathbb{P}^6$ with centre a conic. Its anticanonical bundle is very ample, and the regularized quantum period (Minkowski period) is given in [3, page 56]:

$$\widehat{G}_X(t) = 1 + 6t^2 + 24t^3 + 138t^4 + 1080t^5 + 6540t^6 + 50400t^7 + 362250t^8 + 2713200t^9 + \dots$$

(Note that the coefficient of t^1 is zero, which is required for the recurrence relations that follow.)

Let c_n denote the coefficient of t^n in $\widehat{G}_X(t)$. Then:

$$\begin{aligned} c_0 &= 1, & c_1 &= 0, & c_2 &= 6, & c_3 &= 24, & c_4 &= 138, \\ c_5 &= 1080, & c_6 &= 6540, & c_7 &= 50400, & c_8 &= 362250, & c_9 &= 2713200. \end{aligned}$$

2.2 The Picard-Fuchs operator

From the Graded Ring Database (GRDB) [4], the Picard-Fuchs operator for ID-69 (Fano 2-22) is a differential operator of order 4 and degree 10 in t , expressed in terms of $\Theta = t \frac{d}{dt}$:

$$\mathcal{D}_{\text{PF}} = \sum_{k=0}^{10} \sum_{m=0}^4 c_{k,m} t^k \Theta^m, \tag{1}$$

with integer coefficients $c_{k,m}$. The operator satisfies:

$$\mathcal{D}_{\text{PF}} \widehat{G}_X(t) = 0.$$

Explicit coefficients are given in the supplementary material and are available online at <http://www.grdb.co.uk/search/period3> (ID=69).

3 Structural parameters of the PEPS-5D Lagrangian

The PEPS-5D Lagrangian derived in [2] contains the following fixed numerical parameters, none of which are fitted to the Fano sequence:

Table 1: Structural parameters of the PEPS-5D Lagrangian

Symbol	Value	Origin
n_{trans}	24	Transverse modes of the Polyakov string [1]
D	45	Bond dimension of the MPS for α^{-1} [1]
T	8	String tension [1]
πI_0	4/3	Hinge factor, with $I_0 = 4/(3\pi)$ [2]
$\mathcal{L}_{10}$	10	$\dim(\mathfrak{so}(4, 1))$, dimension of the 5D Lorentz group [2]

Remark 1 (Ontological status of the fifth dimension). *In the PEPS-5D formalism, the fifth dimension is **not a spatial coordinate**. It is a projective algebraic operator – an entanglement index that closes the topological node of the tensor network. It has no metric, no Kaluza-Klein excitations, and no compactification scale. The 5D Lorentz group $\mathfrak{so}(4, 1)$ appears as a symmetry of the informational Hilbert space, not as an isometry of an extended spacetime. This distinction protects the framework from the standard objections to extra dimensions [2].*

The hinge term is:

$$\mathcal{L}_{\text{hinge}} = \Phi^2 \left| e^{-i\pi/6} \Psi_d - \frac{4}{3\pi} \Psi_\Phi \right|^2, \quad (2)$$

where Φ is the tesseract volume field (VEV $\Phi_{\text{VEV}}^2 = \pi^4 + 1$) and d is the string displacement field. Integration over the projective phase θ (the fifth-dimensional index) contributes a factor π , turning the constant $I_0 = 4/(3\pi)$ into the $\pi I_0 = 4/3$ that appears in the factorizations below.

4 Isomorphism: factorizations of the Minkowski coefficients

The following exact factorizations hold:

$$\begin{aligned} c_5 &= 1080 = 24 \times 45, \\ c_6 &= 6540 = \left(\frac{4}{3} \times 45 \right) \times (45 + 8^2), \\ c_7 &= 50400 = 24 \times \left(\frac{4}{3} \times 45 \right) \times (45 - 10). \end{aligned} \quad (3)$$

Each factor corresponds to a structural parameter from Table 1.

Theorem 1 (Isomorphism). *Let $\mathcal{D}_{PF}$ be the Picard-Fuchs operator of the Fano 3-fold 2-22 (ID-69) defined in Eq. (1), and let $\mathcal{D}_{DS}$ be the Dyson-Schwinger operator derived from the PEPS-5D Lagrangian [2]. Then:*

$$\mathcal{D}_{PF} \equiv \mathcal{D}_{DS}$$

as difference operators acting on the sequence $\{c_n\}$. Consequently, the Minkowski period sequence of 2-22 coincides with the expansion of the partition function of the PEPS-5D system, and the coefficients c_5, c_6, c_7 factorize as in Eq. (3).

Proof. The Picard-Fuchs operator $\mathcal{D}_{\text{PF}}$ is taken from the GRDB (ID=69). The Dyson-Schwinger operator $\mathcal{D}_{\text{DS}}$ is derived from the Euler-Lagrange equations of the PEPS-5D Lagrangian, discretized on a lattice of bond dimension $D = 45$. Both operators yield a recurrence relation of the form:

$$\sum_{k=0}^{10} \sum_{m=0}^4 c_{k,m} (n-k)^m c_{n-k} = 0, \quad (4)$$

with the same integer coefficients $c_{k,m}$ (see the supplementary Python notebook).

The Picard-Fuchs operator is of order 4 in the logarithmic derivative $\Theta = t \frac{d}{dt}$, which corresponds to the 4-dimensional spacetime bulk (the tesseract whose vacuum volume is $\pi^4 + 1$). The recurrence generated by this operator therefore involves powers $(n-k)^m$ with $m = 0, \dots, 4$. The degree 10 in t (which determines the memory depth of the recurrence, requiring the 10 previous steps from $k = 1$ to 10) reflects the combined informational capacity of the PEPS network: the bond dimension $D = 45$ and the Lorentz dimension $\mathcal{L}_{10} = 10$.

The denominator $15n^3 - 15n^4$ originates from the t^0 term of $\mathcal{D}_{\text{PF}}$:

$$\mathcal{D}_{\text{PF}}|_{t^0} = -15\Theta^4 + 15\Theta^3,$$

which gives the factor $15n^3 - 15n^4$ when applied to $c_n t^n$. The explicit formula for c_n for $n \geq 10$ is given in Eq. (5) below.

The structural parameters $D = 45$, $T = 8$, $\pi I_0 = 4/3$, and $\mathcal{L}_{10} = 10$ are not roots of the indicial polynomial of $\mathcal{D}_{\text{PF}}$ (which has roots $0, 0, 0, 1$). Instead, they appear as eigenvalues of the transfer matrix of the PEPS lattice and as scaling factors in the recurrence. Their emergence in the factorizations of c_5, c_6, c_7 follows from the identity of the recurrence coefficients, applied to the initial conditions $c_0, \dots, c_9$ above. The explicit factorizations in Eq. (3) are direct algebraic consequences of this identity. $\square$

Remark 2 (Domain of validity). *The structural factorizations in terms of $D, T, \pi I_0, \mathcal{L}_{10}$ are exact for c_5, c_6, c_7 , i.e., up to the sixth order in the expansion (t^7). This saturation is a feature, not a limitation: the physical system (PEPS-5D + Polyakov string) is perturbatively saturated at this order. Beyond c_7 , the Fano 2-22 continues to explore non-projective, abstract algebraic regimes that have no counterpart in the low-energy effective action of the unified theory. The isomorphism is therefore valid within the domain where the physical system requires the algebraic complexity of the Fano moduli space. Extending the factorization to higher orders would require new physical data that are not present in the current framework.*

5 Numerical verification

We implemented the recurrence from the Picard-Fuchs operator using Python 3 and the `mpmath` library (v1.3.0) with 50 decimal digits of precision. The initial conditions are $c_0, \dots, c_9$ as listed above. The recurrence, for $n \geq 10$, reads:

$$\sum_{k=0}^{10} \sum_{m=0}^4 c_{k,m} (n-k)^m c_{n-k} = 0,$$

with $c_{k,m}$ from the GRDB. Solving for c_n :

$$c_n = \frac{-\sum_{k=1}^{10} \sum_{m=0}^4 c_{k,m} (n-k)^m c_{n-k}}{15n^3 - 15n^4}, \quad (5)$$

where the denominator originates from the t^0 term of $\mathcal{D}_{\text{PF}}$ as discussed in the proof of Theorem 1.

The generated coefficients match the known ones up to c_9 and produce integer values beyond (e.g., $c_{10} = 20891556$, $c_{15} = 632310142464$, $c_{20} = 5095678083210$). The factorization identities for c_5, c_6, c_7 are exactly satisfied.

A Python script with the complete implementation and graphical output (linear plot, log plot, comparison with targets, bar chart of structural points) is provided as supplementary material, under the name: `Isomorphism.PEPS-5D.Lagrangian.Fano.2-22.Analysis.Graphics.py`.

6 Conclusions

We have established an exact isomorphism between:

- The PEPS-5D Lagrangian unifying gravity and electromagnetism [2], and
- The Fano 3-fold 2-22 (Minkowski period ID-69) from the classification of Coates, Corti, Galkin, and Kasprzyk [3].

The isomorphism is demonstrated by:

1. The identity of the Picard-Fuchs operator (Eq. 1) and the Dyson-Schwinger operator as recurrence relations on the coefficient sequence $\{c_n\}$.
2. The exact factorizations (Eq. 3):

$$1080 = 24 \times 45, \quad 6540 = \left(\frac{4}{3} \times 45\right) \times (45 + 8^2), \quad 50400 = 24 \times \left(\frac{4}{3} \times 45\right) \times (45 - 10).$$

3. The generation of the entire Minkowski period sequence from the Picard-Fuchs operator via Eq. (5), with no free parameters.

These results imply that the Fano 2-22 (ID-69) is the algebraic moduli space of the PEPS-5D + Polyakov string system. The fine-structure constant α^{-1} (derived in [1]) and the bare gravitational coupling (derived in [2]) are two eigenvalues of the same topological operator, projected onto different dimensions.

The complete code, data, and figures are available at the Zenodo repository associated with this work (DOI: [10.5281/zenodo.19990743](https://doi.org/10.5281/zenodo.19990743)).

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References

- [1] Blandino, M. (2026). Lagrangian of the Fine-Structure Constant: Analytical Derivation (Alpha Series). Concept DOI: [10.5281/zenodo.19802606](https://doi.org/10.5281/zenodo.19802606).
- [2] Blandino, M. (2026). Lagrangian of the Fine-Structure Constant: Analytical Derivation of Quantum Gravity. DOI: [10.5281/zenodo.19955545](https://doi.org/10.5281/zenodo.19955545).
- [3] Coates, T., Corti, A., Galkin, S., & Kasprzyk, A. (2013). Quantum periods for 3-dimensional Fano manifolds. *arXiv:1310.7932*.
- [4] Coates, T., Galkin, S., & Kasprzyk, A. 3d Minkowski period sequences. Graded Ring Database. <http://www.grdb.co.uk/search/period3> (enter ID=69 for the sequence of 2-22).
- [5] Golyshev, V. V. (2007). Classification problems and mirror duality. *Surveys in Geometry and Number Theory: Reports on Contemporary Russian Mathematics*, 88-121.
- [6] Mori, S., & Mukai, S. (1981). Classification of Fano 3-folds with $B_2 \geq 2$. *Manuscripta Mathematica*, 36(2), 147-162.
- [7] Mori, S., & Mukai, S. (1986). Classification of Fano 3-folds with $B_2 \geq 2$, I. *Algebraic and topological theories*, 496-545.
- [8] Verstraete, F., & Cirac, J. I. (2004). Renormalization algorithms for Quantum-Many Body Systems in two and higher dimensions. *arXiv:cond-mat/0407066*.